MAIS PAC

IPhO 02.02: Statics: Forces, Torques and Equilibrium · Lecture + Problems

Statics: Forces, Torques and Equilibrium

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

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1. What equilibrium actually means

An object is in equilibrium when it does not accelerate. For an extended body that requires two vector conditions:

$$\Sigma F = 0 \ \{\text{and}\} \ \Sigma \tau = 0 \ \{\text{about any point.}\} \quad (1)$$

The second condition is what makes statics harder than particle dynamics, because now the **point of application** of each force matters. The torque of a force about a point is

$$\tau = r \times F, \ |\tau| = rF\sin, \quad (2)$$

where is the angle between the position vector and the force. Equivalently, torque is force times perpendicular distance from the pivot to the line of action.

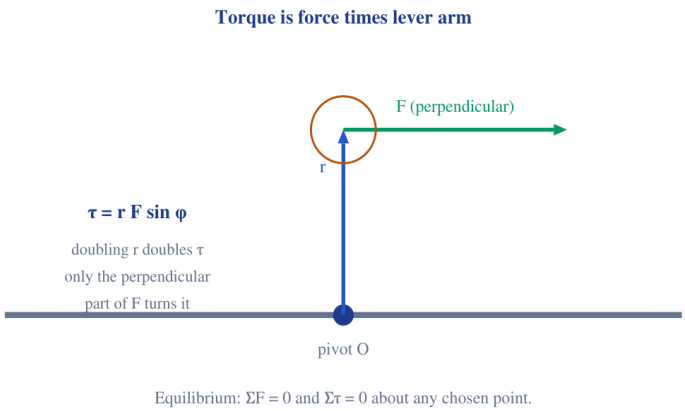


Figure 1. Pick the pivot so that unknown forces pass through it. Their torque is then zero and they vanish from the equation.

Because the choice of pivot is free, choose it cleverly: put it where unknown forces act, so their torque is zero and they disappear from the equation. This is not a trick, it is the reason statics problems are solvable at all.

2. Free-body diagrams that earn marks

A reliable recipe:

1. Isolate **one** body. Draw it as a simple outline.
2. Draw every force **on** it, as an arrow from its point of application: weight at the centre of mass, normal reaction perpendicular to a surface, tension along a string away from the body, friction along the surface opposing relative motion (or the tendency of it).
3. Choose axes. Along a slope, use parallel and perpendicular components, never horizontal and vertical.
4. Write the two force equations and one torque equation.
5. Check that you have as many equations as unknowns.

Common errors that cost marks: putting the normal force at an angle that is not perpendicular to the surface; adding a "centrifugal force" to a static problem; and forgetting the weight of a beam the question said was uniform.

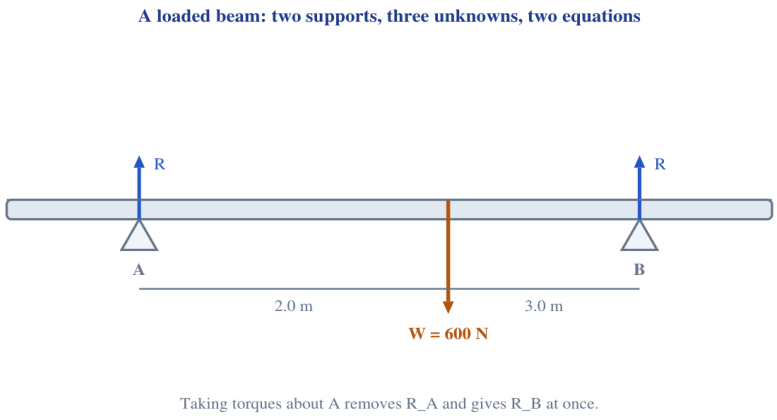


Figure 2. Two force equations and one torque equation are available in two dimensions; a beam with three unknown reactions is exactly determined.

3. Friction: static, kinetic, and the critical angle

Static friction adjusts itself up to a maximum,

$$f_s \mu_s N, \tag{3}$$

and kinetic friction is roughly constant, $f_k = \mu_k N$ with $\mu_k < \mu_s$. The inequality is the whole difficulty: you often have to **test** whether a configuration is possible rather than solve for a number.

On an incline at angle θ, resolving along and perpendicular to the slope gives

$$N = mg\cos\theta,\ f = mg\sin\theta. \tag{4}$$

Sliding begins when $mg\sin\theta = \mu_s mg\cos\theta$, that is

$$\tan\theta_c = \mu_s. \tag{5}$$

The mass cancels. That is why a coin and a truck slide at the same angle on the same surface — and why the result is a clean experimental method for measuring μ_s.

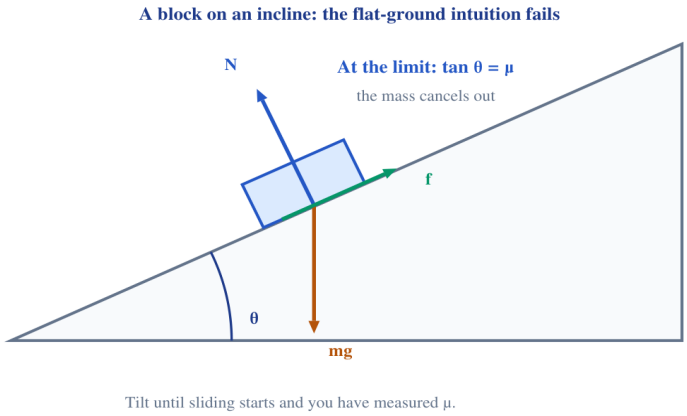


Figure 3. Resolve along and perpendicular to the slope, never horizontally and vertically. The normal reaction is mg cos θ, not mg.

Interactive experiment: A ladder against a smooth wall — open the online handout to use it.

4. Centre of mass, and when a body acts like a point

The centre of mass is the mass-weighted mean position,

$$R = (\Sigma m_i r_i)/(\Sigma m_i) \{or\} R = (1)/(M)\int r, dm. \tag{6}$$

For translation, the whole body behaves as a point mass at R; for rotation, it does not — you need the moment of inertia. For toppling questions, the body is on the edge between stable and unstable when the vertical through its centre of mass passes through the pivot point.

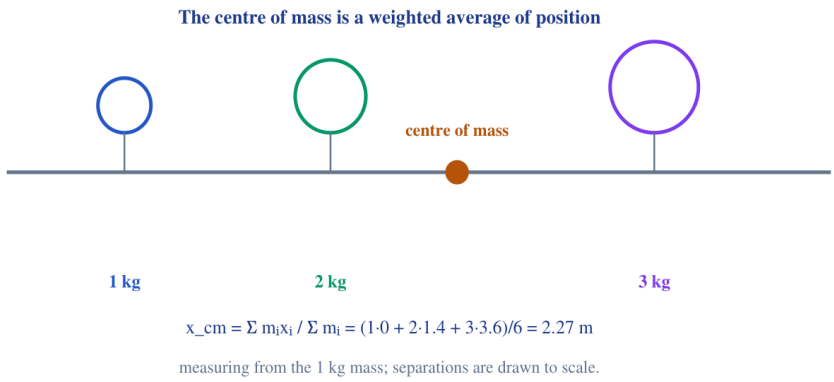


Figure 5. A rigid body acts like a point mass at its centre of mass for translation; for rotation, the distribution matters and you need the moment of inertia.

5. Stability: the potential tells you everything

Equilibrium is **stable** if a small displacement produces a restoring force, **unstable** if it grows, and **neutral** if nothing happens. In energy language, expand the potential about an equilibrium x_0:

$$U(x) = U(x_0) + \{U'(x_0)\}_{(=0)}(x-x_0) + (1)/(2)\, U''(x_0)(x-x_0)^2 + \qquad (7)$$

- U'' > 0: a parabolic well, stable, with ω = √(U''/m).
- U'' < 0: a maximum, unstable, displacement grows exponentially.
- U'' = 0: inconclusive from this order; look at the next term.

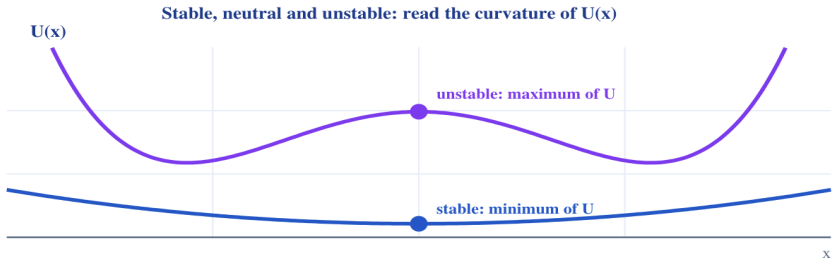


Figure 4. Expand U about an equilibrium: the linear term vanishes, the quadratic term decides everything, and ω² = U″/m.

The statics checklist. Free-body diagram with every force at its true point of application → axes chosen along the geometry → force equations → torque equation about the cleverest pivot → check the number of unknowns → test the friction inequality explicitly, and say which way the body would move if it failed.

6. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — A balanced rule. A uniform metre rule of mass 120 {g} is pivoted at the 50 {cm} mark. A 200 {g} mass hangs at the 20 {cm} mark. (a) Where must a 150 {g} mass hang for balance? (b) Find the reaction at the pivot. (c) Explain what changes if the pivot is moved to the 30 {cm} mark.

Worked solution →

[1] Problem 2 — The ladder problem. A uniform ladder of length L and mass m leans against a smooth vertical wall at angle θ to the horizontal. The floor is rough with coefficient μ. (a) Find the normal reaction at the wall and at the floor. (b) Find the friction the floor must supply. (c) Show that the ladder slips when $\tan\theta < 1/(2\mu)$, and compute the critical angle for $\mu = 0.45$.

Worked solution →

[2] Problem 3 — Beam with two loads. A uniform beam of mass 40 {kg} and length 6.0 {m} rests on supports at its ends. A 60 {kg} mass sits 1.5 {m} from the left end and a 30 {kg} mass sits 4.5 {m} from the left end. (a) Find both support reactions. (b) Find where a single 90 {kg} mass would have to sit to produce the same reactions. (c) Explain why the reactions change if the beam is not uniform but the total mass is unchanged.

Worked solution →

[2] Problem 4 — Three forces, one triangle. A body is in equilibrium under exactly three forces. (a) Prove that the three force vectors must form a closed triangle when drawn head to tail. (b) Prove that their lines of action must be concurrent. (c) Use this to find the tension in each of two strings supporting a lamp of weight W if the strings make angles 40° and 65° with the ceiling.

Worked solution →

[2] Problem 5 — Sliding versus toppling. A uniform block of height h and width w stands on a rough horizontal floor. A horizontal force is applied at height y above the floor. (a) Find the value of y at which sliding and toppling occur simultaneously. (b) Show that for y above this value the block topples before it slides. (c) A tall bookshelf of height 1.8 {m} and width 0.35 {m} stands on a floor with $\mu = 0.5$: does it slide or topple when pushed at the top?

Worked solution →

[2] Problem 6 — Hanging by a thread: the sagging rope. A rope of negligible mass carries a 25 {kg} load at its midpoint. Its two halves make angles θ with the horizontal, and the rope can withstand a tension of 1200 {N}. (a) Find the tension as a function of θ. (b) Find the smallest angle that is safe. (c) Explain why a nearly horizontal rope cannot hold a load, however strong it is.

Worked solution →

[2] Problem 7 — Contact forces between stacked blocks. Three identical blocks of mass m are stacked, and a horizontal force F pushes the bottom block. The coefficient of friction between all surfaces is μ. (a) Find the maximum acceleration for which the upper two blocks do not slip. (b) Determine which pair slips first as F increases. (c) Find the acceleration just before slip and the force required.

Worked solution →

[2] Problem 8 — Centre of mass of a composite shape. A uniform lamina is made of a square of side 2a with a circular hole of radius a/2 removed, the hole centred at the midpoint of the right-hand edge. (a) Find the centre of mass of the remaining lamina, measuring from the centre of the square. (b) Find the centre of mass if instead a half-disc of radius a is added along the right edge. (c) Verify both answers with a limiting case.

Worked solution →

[2] Problem 9 — Small oscillations from a potential. A particle of mass m moves in the potential U(x) = U_0((a)/(x) - (a²)/(2x²)) with U_0, a > 0 and x>0. (a) Find the equilibrium position. (b) Show that it is stable and find the oscillation frequency for small displacements. (c) Show that U(x) is bounded above, and interpret the motion when the particle's total energy exceeds that bound.

Worked solution →

[3] Problem 10 — A tripod and a three-dimensional equilibrium. A camera of weight W is on a tripod whose three legs each make an angle α with the vertical and are equally spaced in azimuth. (a) Show by symmetry that the vertical force on each leg is W/3. (b) Show that the legs exert a net inward horizontal force on the head, and find its magnitude per leg. (c) Explain why the legs must splay outward, and find the minimum friction coefficient with the ground that keeps them from sliding if friction at the head is absent.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official

marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

HyperPhysics mechanics map

A compact concept map for revising definitions and standard results quickly.